



Performance, Load & Stress Benchmark Scenarios

Concurrency Benchmarks, High-Volume POS Latency & Query Profiling

DOCUMENT TYPE	TARGET AUDIENCE	RELEASE VERSION	STATUS
Performance Benchmark	Performance Engineers, Tech Leads	v1.4.0 Production	✓ Verified & Approved

Performance Test Scenarios

This document outlines detailed performance test scenarios and benchmarks for the StockWhisk SaaS project.

Load Testing

TEST ID	SCENARIO	TARGET METRIC	NOTES
PT-LOAD-001	50 concurrent POS checkout transactions	< 2s response time	Simulates peak hour retail checkout volume across multiple tenants.
PT-LOAD-002	100 concurrent product searches	< 500ms response time	Focuses on database read replica and search indexing performance.
PT-LOAD-003	Dashboard load with 10,000+ sales records	< 3s response time	Tests aggregation queries (SUM, COUNT) for daily/monthly metrics.
PT-LOAD-004	Report generation with 1 year of data	< 5s response time	Tests heavy background processing or optimized materialized views.
PT-LOAD-005	Barcode scan response	< 200ms response time	CRITICAL for POS UX; requires minimal latency.

Stress Testing

TEST ID	SCENARIO	EXPECTED BEHAVIOR UNDER STRESS
PT-STRESS-001	200 concurrent users on POS	Graceful degradation; no data corruption; latency increases but requests don't drop.
PT-STRESS-002	Bulk product import (1000+ items) via CSV	Background task queue handles load; UI remains responsive; import completes within 1 minute.
PT-STRESS-003	Database query spike	Connection pool handles requests without dropping; slow queries logged for review.

Scalability

TEST ID	SCENARIO	ARCHITECTURE FOCUS
PT-SCALE-001	System handles 100+ active shops	Multi-tenant routing efficiency; global cache hit rates.
PT-SCALE-002	Single shop with 50,000+ products	Indexing on <code>tenant_id</code> + <code>product_id</code> ; pagination efficiency.
PT-SCALE-003	Single shop with 100,000+ invoices	Date-range query efficiency; storage cost optimization for PDF generation.

Response Time Benchmarks

The following targets must be maintained under standard load:

- **POS Barcode Scan:** < 200ms
- **Checkout Completion (Sale save):** < 2s
- **Dashboard Initial Load:** < 3s
- **Custom Report Generation:** < 5s
- **Product Search (Typeahead):** < 500ms
- **Invoice PDF Generation & Download:** < 3s

Database Performance Requirements

- **Index Verification:** Ensure B-Tree indexes exist on all foreign keys and frequently searched columns (e.g., `barcode` , `phone` , `invoice_number`).
- **N+1 Query Prevention:** Utilize Django's `select_related` and `prefetch_related` on all listing APIs to strictly cap the query count per request (target < 5 queries per complex list).
- **Connection Pooling:** Verify PgBouncer (or similar) is configured correctly to prevent connection exhaustion during peak load.